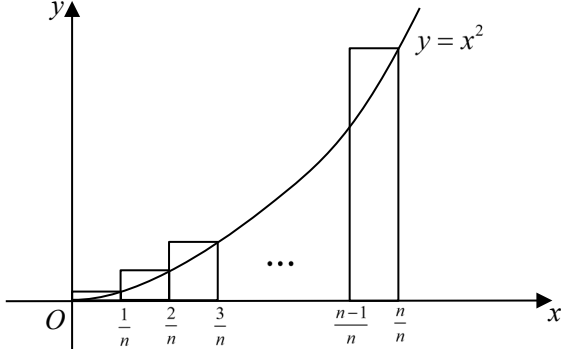
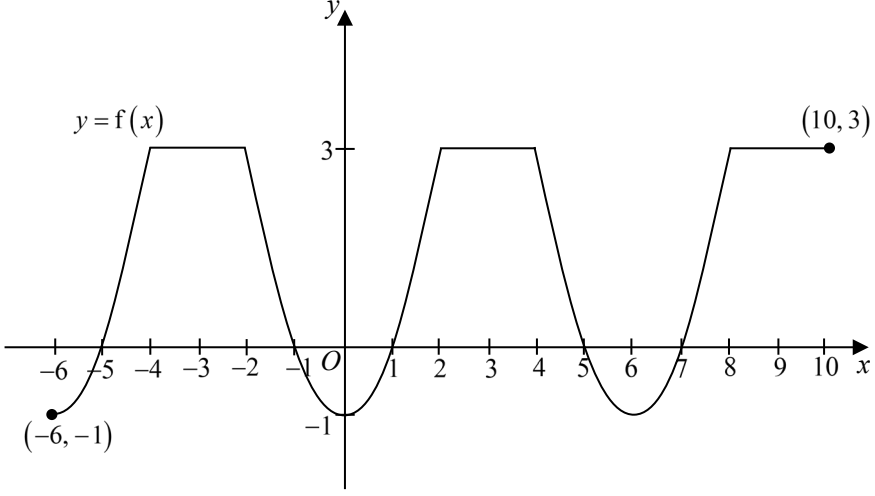
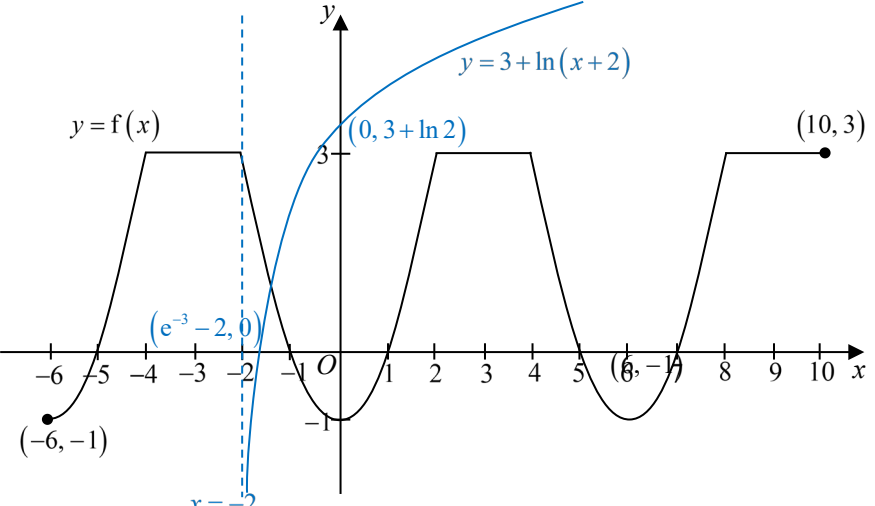
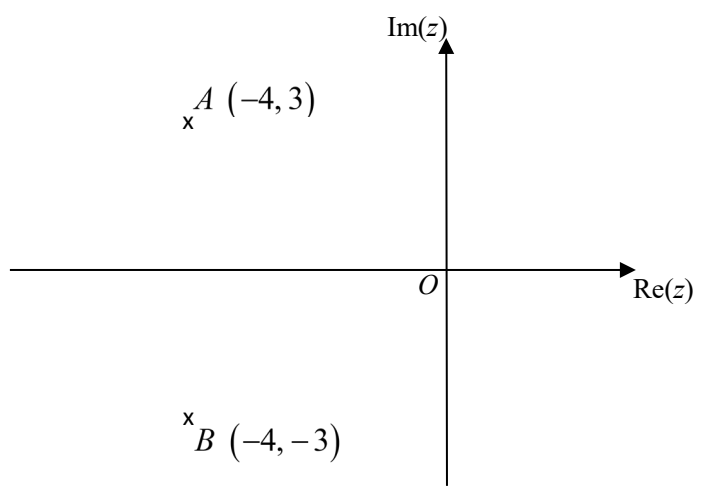


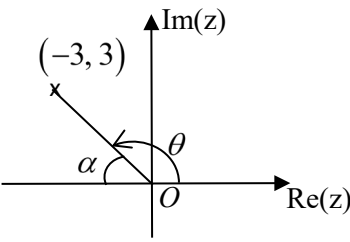
MI 2020 PU3 H2 MATHS 9758/02 PRELIM EXAM SOLUTION

Section A: Pure Mathematics

Qn	Solution
1(i)	 Width of each rectangle $= \frac{1}{n}$ Area of 1st rectangle $= \frac{1}{n} \times \left(\frac{1}{n}\right)^2$ Area of 2nd rectangle $= \frac{1}{n} \times \left(\frac{2}{n}\right)^2$... Area of nth rectangle $= \frac{1}{n} \times \left(\frac{n}{n}\right)^2$ Total area of n rectangles $= \frac{1}{n} \times \left(\frac{1}{n}\right)^2 + \frac{1}{n} \times \left(\frac{2}{n}\right)^2 + \dots + \frac{1}{n} \times \left(\frac{n}{n}\right)^2$ $= \sum_{r=1}^n \frac{1}{n} \times \left(\frac{r}{n}\right)^2$ $= \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^2$ $= \frac{1}{n^3} \sum_{r=1}^n r^2 \text{ (shown)}$
1(ii)	Hence $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^2 = \text{area under the graph of } y = x^2 \text{ from } x = 0 \text{ to } x = 1$ $= \int_0^1 x^2 \, dx$ $= \left[\frac{x^3}{3} \right]_0^1$ $= \frac{1}{3}$

Qn	Solution
2(i)	
2(ii)	To be drawn in same diagram as (i) 
2(iii)	Using GC, the x-coordinate of the point of intersection between $y = f(x)$ and $y = 3 + \ln(x + 2)$ is -1.6859. Hence, from the graph above, the solution to $f(x) \geq 3 + \ln(x - 3)$ is $-2 < x \leq -1.69 \text{ (3sf)}$

Qn	Solution
3(a) First part	Method 1: Using Quadratic Formula $z = \frac{-8 \pm \sqrt{8^2 - 4(1)(25)}}{2}$ $= \frac{-8 \pm \sqrt{-36}}{2}$ $= \frac{-8 \pm 6i}{2}$ $= -4 \pm 3i$ ----- Method 2: Completing the square $z^2 + 8z + 25 = (z + 4)^2 + 9$ $(z + 4)^2 + 9 = 0$ $\Rightarrow (z + 4)^2 = -9$ $\Rightarrow z + 4 = \pm \sqrt{-9}$ $= \pm 3i$ $\Rightarrow z = -4 \pm 3i$ ----- Since $\arg(z_1) > 0$, $z_1 = -4 + 3i$ $z_2 = -4 - 3i$ Let A and B be the points representing z_1 and z_2 on the Argand diagram respectively. 
3(a) 2nd part	$z_1 + 1 = -3 + 3i$ $ -3 + 3i = \sqrt{(-3)^2 + 3^2}$ $= \sqrt{18}$ $= 3\sqrt{2}$

Qn	Solution
	$\arg(-3+3i) = \pi - \alpha$ $= \pi - \tan^{-1}\left(\frac{3}{3}\right)$ $= \pi - \frac{\pi}{4}$ $= \frac{3\pi}{4}$ 
3(b)	Using triangle inequality, $ 1+ic + c+i \geq 1+ic+c+i $ $= (1+c)+i(1+c) $ $\sqrt{1+c^2} + \sqrt{1+c^2} \geq \sqrt{(1+c)^2 + (1+c)^2}$ $2\sqrt{1+c^2} \geq \sqrt{2(1+c)^2}$ Since c is a real and positive constant, $1+c > 0$. $2\sqrt{1+c^2} \geq \sqrt{2} (1+c)$ $\sqrt{2}\sqrt{1+c^2} \geq 1+c$ $\therefore 1+c \leq \sqrt{2(1+c^2)} \quad (\text{shown})$
4(a) (i)	Method 1: When $p=1$ and $q = \frac{\pi}{2}$, $x = a \sin\left(t + \frac{\pi}{2}\right)$ $\frac{dx}{dt} = a \cos\left(t + \frac{\pi}{2}\right)$ $y = b \sin t$ $\frac{dy}{dt} = b \cos t$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ $= \frac{b \cos t}{a \cos\left(t + \frac{\pi}{2}\right)}$ $= \frac{b \cos t}{a \left(\cos t \cos \frac{\pi}{2} - \sin t \sin \frac{\pi}{2}\right)}$ $= \frac{b \cos t}{-a \sin t}$ $= -\frac{b}{a} \cot t \quad (\text{shown})$

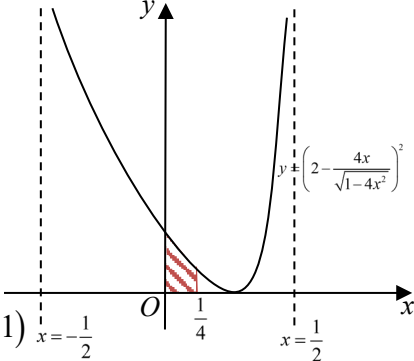
Qn	Solution
	----- Method 2: $x = a \sin\left(t + \frac{\pi}{2}\right)$ $= a \left(\sin t \cos \frac{\pi}{2} + \cos t \sin \frac{\pi}{2} \right)$ $= a \cos t$ $\frac{dx}{dt} = -a \sin t$ $y = b \sin t$ $\frac{dy}{dt} = b \cos t$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ $= \frac{b \cos t}{-a \sin t}$ $= -\frac{b}{a} \cot t \text{ (shown)}$
4(a) (ii)	At $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$, Method 1: From $x = \frac{a\sqrt{3}}{2}$, $a \sin\left(t + \frac{\pi}{2}\right) = \frac{a\sqrt{3}}{2}$ $\Rightarrow \sin\left(t + \frac{\pi}{2}\right) = \frac{\sqrt{3}}{2}$ $t + \frac{\pi}{2} = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$ $t = -\frac{\pi}{6} \text{ or } \frac{\pi}{6}$ From $y = \frac{b}{2}$, $b \sin t = \frac{b}{2} \Rightarrow \sin t = \frac{1}{2}$ $t = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$ Method 2: $x = a \sin\left(t + \frac{\pi}{2}\right)$ $= a \left(\sin t \cos \frac{\pi}{2} + \cos t \sin \frac{\pi}{2} \right)$ $= a \cos t$ From $x = \frac{a\sqrt{3}}{2}$, $a \cos t = \frac{a\sqrt{3}}{2} \Rightarrow \cos t = \frac{\sqrt{3}}{2}$ $t = -\frac{\pi}{6} \text{ or } \frac{\pi}{6}$

Qn	Solution
	$\therefore t = \frac{\pi}{6}$ ----- Method 3: From $y = \frac{b}{2}$, $b \sin t = \frac{b}{2} \Rightarrow \sin t = \frac{1}{2} \text{ --- (1)}$ From $x = a \sin\left(t + \frac{\pi}{2}\right)$, $a \sin\left(t + \frac{\pi}{2}\right) = \frac{a\sqrt{3}}{2} \Rightarrow \sin\left(t + \frac{\pi}{2}\right) = \frac{\sqrt{3}}{2}$ $\Rightarrow \sin t \cos \frac{\pi}{2} + \cos t \sin \frac{\pi}{2} = \frac{\sqrt{3}}{2}$ $\Rightarrow \cos t = \frac{\sqrt{3}}{2} \text{ --- (2)}$ $\frac{(1)}{(2)}: \frac{\sin t}{\cos t} = \frac{1/2}{\sqrt{3}/2}$ $\Rightarrow \tan t = \frac{1}{\sqrt{3}}$ $\Rightarrow t = \frac{\pi}{6} \text{ or } -\frac{5\pi}{6} \text{ (rej as } \cos\left(-\frac{5\pi}{6}\right) \neq \frac{\sqrt{3}}{2})$
4(a) (iii)	Method 1: At $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$, $\frac{dy}{dx} = -\frac{1}{2/\sqrt{3}} = -\frac{\sqrt{3}}{2}$ Using $\frac{dy}{dx} = -\frac{b}{a} \cot t$ at $t = \frac{\pi}{6}$, $-\frac{b}{a} \cot \frac{\pi}{6} = -\frac{\sqrt{3}}{2}$ $-\frac{b}{a} \frac{1}{\tan \frac{\pi}{6}} = -\frac{\sqrt{3}}{2}$ $-\frac{b}{a} (\sqrt{3}) = -\frac{\sqrt{3}}{2}$ $\frac{b}{a} = \frac{1}{2}$ $2b = a \text{ --- (1)}$

Qn	Solution
	Since the point $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$ lies on the normal, $\frac{b}{2} = \frac{2}{\sqrt{3}}\left(\frac{a\sqrt{3}}{2}\right) - \frac{3}{2}$ $\frac{b}{2} = a - \frac{3}{2}$ $b = 2a - 3 \quad \text{--- (2)}$ Substitute (1) into (2): $b = 2(2b) - 3$ $\Rightarrow 3b = 3$ $\Rightarrow b = 1$ Substitute $b = 1$ into (1): $a = 2$ ----- Method 2: At $t = \frac{\pi}{6}$, $\frac{dy}{dx} = -\frac{b}{a} \cot\left(\frac{\pi}{6}\right)$ $= -\frac{b}{a} \frac{1}{\tan\left(\frac{\pi}{6}\right)}$ $= -\frac{b}{a}(\sqrt{3})$ Gradient of normal at $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right) = -\frac{1}{-\frac{b}{a}(\sqrt{3})}$ $= \frac{a}{b\sqrt{3}}$ Equation of normal at the point $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$: $y - \frac{b}{2} = \frac{a}{\sqrt{3}b}\left(x - \frac{a\sqrt{3}}{2}\right)$ $y = \left(\frac{a}{\sqrt{3}b}\right)x + \frac{b}{2} - \frac{a^2}{2b}$

Qn	Solution
	By comparison to $y = \frac{2}{\sqrt{3}}x - \frac{3}{2}$, $\frac{a}{\sqrt{3}b} = \frac{2}{\sqrt{3}} \Rightarrow a = 2b \text{ --- (1)}$ $\frac{b}{2} - \frac{a^2}{2b} = -\frac{3}{2} \Rightarrow b - \frac{a^2}{b} = -3 \text{ --- (2)}$ Substitute (1) into (2): $b - \frac{4b^2}{b} = -3$ $\Rightarrow b - 4b = -3$ $\Rightarrow b = 1$ Substitute $b = 1$ into (1): $a = 2$
4(b)	<u>Method 1:</u> When $p = 1$ and $q = 0$, $x = a \sin t$ $y = b \sin t$ $\frac{x}{y} = \frac{a}{b}$ $y = \frac{b}{a}x$ Hence C is a straight line. ----- <u>Method 2:</u> When $p = 1$ and $q = 0$, $x = a \sin t$ $y = b \sin t$ $\frac{dx}{dt} = a \cos t, \quad \frac{dy}{dt} = b \cos t$ $\frac{dy}{dx} = \frac{b \cos t}{a \cos t} = \frac{b}{a}$ Since the gradient is a constant, C is a straight line.
5(i)	Method 1: Use $\int \frac{1}{a^2 - x^2} dx$ formula in MF26

Qn	Solution
	$\int \frac{1}{1-4x^2} dx = \int \frac{1}{4\left(\frac{1}{4}-x^2\right)} dx$ $= \frac{1}{4} \int \frac{1}{\frac{1}{4}-x^2} dx$ $= \frac{1}{4} \left(\frac{1}{2\left(\frac{1}{2}\right)} \right) \ln \left \frac{\frac{1}{2}+x}{\frac{1}{2}-x} \right + c$ $= \frac{1}{4} \ln \left \frac{1+2x}{1-2x} \right + c$ ----- Method 2: Partial Fractions Using partial fractions, Let $\frac{1}{1-4x^2} = \frac{A}{1-2x} + \frac{B}{1+2x}$ $1 = A(1+2x) + B(1-2x)$ Comparing constant term, $A + B = 1$ ---- (1) Comparing x term, $2A - 2B = 0$ ---- (2) Solving simultaneously, $A = B = \frac{1}{2}$ $\int \frac{1}{1-4x^2} dx = \int \frac{1}{2(1-2x)} + \frac{1}{2(1+2x)} dx$ $= \frac{1}{2} \int \frac{1}{1-2x} dx + \frac{1}{2} \int \frac{1}{1+2x} dx$ $= \frac{1}{2} \left(-\frac{1}{2} \ln 1-2x + \frac{1}{2} \ln 1+2x \right) + c$ $= \frac{1}{4} \ln \left \frac{1+2x}{1-2x} \right + c$

Qn	Solution
5(ii)	$\int \frac{x}{\sqrt{1-4x^2}} dx = -\frac{1}{8} \int \frac{-8x}{\sqrt{1-4x^2}} dx$ $= -\frac{1}{8} \left[\frac{(1-4x^2)^{\frac{1}{2}}}{\frac{1}{2}} \right] + c$ $= -\frac{1}{4} \sqrt{1-4x^2} + c$
5(iii)	$\int \left(2 - \frac{4x}{\sqrt{1-4x^2}} \right)^2 dx$ $= \int 4 - \frac{16x}{\sqrt{1-4x^2}} + \frac{16x^2}{1-4x^2} dx$ $= \int 4 - \frac{16x}{\sqrt{1-4x^2}} - 4 + \frac{4}{1-4x^2} dx$ $= \int -\frac{16x}{\sqrt{1-4x^2}} + \frac{4}{1-4x^2} dx$ $= -16 \int \frac{x}{\sqrt{1-4x^2}} dx + 4 \int \frac{1}{1-4x^2} dx$ $= -16 \left(-\frac{1}{4} \sqrt{1-4x^2} \right) + 4 \left(\frac{1}{4} \ln \left \frac{1+2x}{1-2x} \right \right) + c$ $= 4\sqrt{1-4x^2} + \ln \left \frac{1+2x}{1-2x} \right + c$
	$\text{Area} = \int_0^{\frac{1}{4}} \left(2 - \frac{4x}{\sqrt{1-4x^2}} \right)^2 dx$ $= \left[4\sqrt{1-4x^2} + \ln \left \frac{1+2x}{1-2x} \right \right]_0^{\frac{1}{4}}$ $= \left(4\sqrt{1-\frac{1}{4}} + \ln \left \frac{\frac{3}{2}}{\frac{1}{2}} \right \right) - (4 + \ln 1)$ $= 2\sqrt{3} + \ln 3 - 4 \text{ units}^2$ 

Section B: Probability and Statistics

Qn	Solution																																																								
6(i)	No. of ways = $\frac{9!}{2!2!} = 90720$																																																								
6(ii)	<u>Case 1: Begin and end with ‘T’</u> No. of ways = $4! \times {}^5C_3 \times \frac{3!}{2!} = 720$ <u>Case 2: Begin with ‘T’ but not end with ‘T’ (vice versa)</u> No. of ways = ${}^4C_1 \times 4! \times {}^5C_3 \times \frac{3!}{2!} \times 2! = 5760$ <u>Case 3: Begin and end with non ‘T’</u> No. of ways = ${}^4C_2 \times 2! \times \frac{4!}{2!} \times {}^5C_3 \times \frac{3!}{2!} = 4320$ Total number of ways = 10800																																																								
7(i)	Total probability = 1 $k(1)^2 + \frac{1}{3} + k(2)^2 + \frac{1}{3} + 3k + 4k = 1$ $12k = \frac{1}{3}$ $k = \frac{1}{36}$																																																								
7(ii)	Let T be the score on the biased four-sided die and S be the score on the fair six-sided die. <table border="1"><thead><tr><th colspan="2"></th><th colspan="6">S: Fair (Prob = 1/6)</th></tr><tr><th>T:</th><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr></thead><tbody><tr><td>Prob</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>13/36</td><td>1</td><td>0</td><td>4</td><td>4</td><td>4</td><td>4</td><td>4</td></tr><tr><td>16/36</td><td>2</td><td>−6</td><td>0</td><td>4</td><td>4</td><td>4</td><td>4</td></tr><tr><td>3/36</td><td>3</td><td>−6</td><td>4</td><td>0</td><td>4</td><td>4</td><td>4</td></tr><tr><td>4/36</td><td>4</td><td>−6</td><td>−6</td><td>4</td><td>0</td><td>4</td><td>4</td></tr></tbody></table> $\begin{aligned} P(X = -6) &= P(T = 2, S = 1) + P(T = 3, S = 1) \\ &\quad + P(T = 4, S = 1) + P(T = 4, S = 2) \\ &= \frac{1}{6} \left[\left(\frac{1}{36}(2)^2 + \frac{1}{3} \right) + \frac{1}{36}(3) + \frac{1}{36}(4)(2) \right] \\ &= \frac{1}{8} \end{aligned}$			S: Fair (Prob = 1/6)						T:		1	2	3	4	5	6	Prob								13/36	1	0	4	4	4	4	4	16/36	2	−6	0	4	4	4	4	3/36	3	−6	4	0	4	4	4	4/36	4	−6	−6	4	0	4	4
		S: Fair (Prob = 1/6)																																																							
T:		1	2	3	4	5	6																																																		
Prob																																																									
13/36	1	0	4	4	4	4	4																																																		
16/36	2	−6	0	4	4	4	4																																																		
3/36	3	−6	4	0	4	4	4																																																		
4/36	4	−6	−6	4	0	4	4																																																		

$$\begin{aligned}
 P(X=0) &= P(T=1, S=1) + P(T=2, S=2) \\
 &\quad + P(T=3, S=3) + P(T=4, S=4) \\
 &= \frac{1}{6} \left[\left(\frac{1}{36}(1)^2 + \frac{1}{3} \right) + \left(\frac{1}{36}(2)^2 + \frac{1}{3} \right) + \frac{1}{36}(3) + \frac{1}{36}(4) \right] \\
 &= \frac{1}{6}
 \end{aligned}$$

Method 1 (To find $P(X=4)$)

$$\begin{aligned}
 P(X=4) &= 1 - P(X=-6) - P(X=0) \\
 &= \frac{17}{24}
 \end{aligned}$$

Method 2 (To find $P(X=4)$)

$$\begin{aligned}
 P(X=4) &= P(T=1, S=2, 3, 4, 5, 6) + P(T=2, S=3, 4, 5, 6) \\
 &\quad + P(T=3, S=2, 4, 5, 6) + P(T=4, S=3, 5, 6) \\
 &= \left(\frac{1}{6} \right) \left[5 \left(\frac{1}{36}(1)^2 + \frac{1}{3} \right) + 4 \left(\frac{1}{36}(2)^2 + \frac{1}{3} \right) \right. \\
 &\quad \left. + 4 \left(\frac{1}{36}(3) \right) + 3 \left(\frac{1}{36}(4) \right) \right] \\
 &= \frac{17}{24}
 \end{aligned}$$

x	-6	0	4
$P(X=x)$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{17}{24}$

7(iii) $E(\text{Brandon winning}) = -6 \left(\frac{1}{8} \right) + 4 \left(\frac{17}{24} \right)$

$$= \$ 2.08 > 0$$

Thus the game is in favour of Brandon.

8(a)

(i) Method 1:

$$P = \frac{12}{12} \times \frac{11}{12} \times \frac{10}{12} = \frac{55}{72}$$

Method 2:

$$P = \frac{{}^{12}C_3 \times 3!}{12^3} \text{ or } \frac{{}^{12}P_3}{12^3}$$

$$= \frac{55}{72}$$

8(a) (ii)	Method 1: When $n = 4$, $P = \frac{55}{72} \times \frac{9}{12} = 0.57292 \text{ or } \frac{55}{96}$ When $n = 5$, $P = \frac{275}{432} \times \frac{8}{12} = 0.42438 \text{ or } \frac{275}{648}$ Hence smallest value of n is 5. ----- Method 2: When $n = 4$, $P = \frac{{}^{12}C_4 \times 4!}{12^4} \text{ or } \frac{{}^{12}P_4}{12^4}$ $= \frac{55}{96}$ For n people, $P = \frac{{}^{12}C_n \times n!}{12^n} \text{ or } \frac{{}^{12}P_n}{12^n}$ $\frac{{}^{12}C_n \times n!}{12^n} < \frac{1}{2}$ Using GC, <table border="1" data-bbox="231 1070 461 1312"> <thead> <tr> <th>n</th><th>$\frac{{}^{12}C_n \times n!}{12^n}$</th></tr> </thead> <tbody> <tr> <td>4</td><td>0.5729</td></tr> <tr> <td>5</td><td>0.3819</td></tr> <tr> <td>6</td><td>0.2228</td></tr> <tr> <td></td><td></td></tr> </tbody> </table> Hence smallest value of n is 5.	n	$\frac{{}^{12}C_n \times n!}{12^n}$	4	0.5729	5	0.3819	6	0.2228		
n	$\frac{{}^{12}C_n \times n!}{12^n}$										
4	0.5729										
5	0.3819										
6	0.2228										
8(a) (iii)	Least possible value of n is 13. Since there are only 12 cards in the pack, at least 2 of the people will definitely draw the same card if there are at least 13 people in the group.										
8(b)	$P(\text{at least 2 people picked the same card}) > \frac{1}{2}$ $\Rightarrow 1 - P(\text{all the people picked different cards}) > \frac{1}{2}$ $\Rightarrow P(\text{all the people picked different cards}) < \frac{1}{2}$ ----- Method 1: When $n = 13$, $P(\text{all picked diff cards}) = 0.56640 \times \frac{108}{120} = 0.50976$ When $n = 14$,										

$$P(\text{all picked diff cards}) = 0.50976 \times \frac{107}{120} = 0.45453$$

Hence the smallest value of n is 14

Method 2:

For n people,

$$P(\text{all picked diff cards}) = \frac{{}^{120}C_n \times n!}{120^n} \text{ or } \frac{{}^{120}P_n}{120^n}$$

$$\frac{{}^{120}C_n \times n!}{120^n} < \frac{1}{2}$$

Using GC,

n	$\frac{{}^{120}C_n \times n!}{120^n}$
13	0.5098
14	0.4545
15	0.4015

Hence smallest value of n is 14.

- 9(i)** (1) The probability that a randomly chosen reusable ice pack gel leaks is constant.
- (2) The event that a randomly chosen reusable ice pack gel leaks is independent of another randomly chosen reusable ice pack gel

9(ii)

Let a be $\frac{p}{100}$

Since the most likely number is 3,

$$P(X=2) < P(X=3)$$

$${}^{120}C_2 \times a^2 \times (1-a)^{118} < {}^{120}C_3 \times a^3 \times (1-a)^{117}$$

$$1-a < \frac{118}{3}a$$

$$a > \frac{3}{121}$$

$$a > 0.0248 \text{ (3 s.f.)}$$

AND

$$P(X=3) > P(X=4)$$

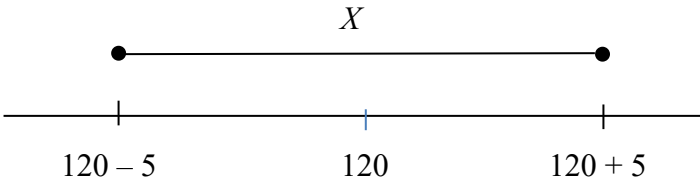
$${}^{120}C_3 \times a^3 \times (1-a)^{117} > {}^{120}C_4 \times a^4 \times (1-a)^{116}$$

$$1-a > \frac{117}{4}a$$

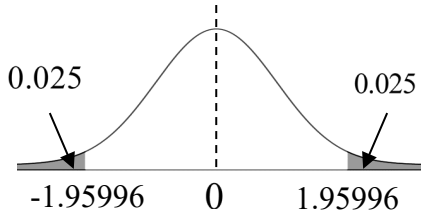
$$a < \frac{4}{121}$$

$$a < 0.0331 \text{ (3 s.f.)}$$

$$\therefore 2.48 < p < 3.31$$

9(iii)	$X \sim B(120, 0.03)$ $P(X \leq 4) \approx 0.70763 = 0.708 \text{ (3 s.f.)}$
9(iv)	$P(X < 9 X \geq 5) = \frac{P(5 \leq X \leq 8)}{P(X \geq 5)}$ $= \frac{P(X \leq 8) - P(X \leq 4)}{1 - P(X \leq 4)}$ $\approx 0.96434 = 0.964 \text{ (3 s.f.)}$
9(v)	Let Y be the random variable, the number of days with at most 4 reusable ice pack gel that leaks (out of 7) $Y \sim B(7, 0.70763)$
	Required Prob $= P(Y = 2) \times 0.70763 \approx 0.015897 = 0.0159 \text{ (3 s.f.)}$
10(i)	Let X be the mass (in grams) of a randomly chosen orange $X \sim N(120, 16^2)$
	 X 120 - 5 120 120 + 5 <u>Method 1</u> $P(115 \leq X \leq 125) \approx 0.24534 = 0.245 \text{ (3 s.f.)}$ <u>Method 2</u> $P(X - 120 \leq 5) = P(115 \leq X \leq 125) \approx 0.24534$ $= 0.245 \text{ (3 s.f.)}$
10(ii)	Let Y be the mass (in grams) of a randomly chosen carrot $Y \sim N(73, 6^2)$ $P(Y > k) < 0.05$ Critical value, $k = 82.869$ Set of values of $k = \{k \in \mathbb{R} : k > 82.9\}$
10(iii)	Let $A = X_1 + X_2 + \dots + X_4 - (Y_1 + Y_2 + \dots + Y_7)$ $A \sim N(-31, 1276)$

	$P(X_1 + X_2 + \dots + X_4 - (Y_1 + Y_2 + \dots + Y_7) \geq 5)$ $= P(A \geq 5)$ $\approx 0.15677 = 0.157 \text{ (3 s.f.)}$
10 (iv)	Let $C = 0.7(X_1 + X_2 + \dots + X_8) + 0.88(Y_1 + Y_2 + \dots + Y_{14})$ $C \sim N(1571.36, 1393.8176)$ $P(C < 1600)$ $\approx 0.77850 = 0.779 \text{ (3 s.f.)}$
10 (v)	The masses of randomly chosen oranges and carrots are independent of each other.
11(i)	The 8 classes of Primary Six students form a sample as they are observations taken from the Nurturing Primary School population.
11 (ii)	Mrs Neha can number the students using the class list from the first class in Primary One to the last class in Primary Six. She can use a random number generator to generate a random sample of 100 students. By conducting the survey in this manner, every student in the school will have an equal chance of being selected and the selection of one student will be independent of another.
11 (iii) (a)	Unbiased estimate of the population mean, $\bar{x} = 30 + \frac{\sum(x-30)}{100} = 29.57$ (exact) Unbiased estimate of the population variance, $s^2 = \frac{1}{99} \left[911 - \frac{(-43)^2}{100} \right] = \frac{89251}{9900} \approx 9.01525 = 9.02 \text{ (3 s.f.)}$
11 (iii) (b)	Let X be the screen time per day, in minutes, of a randomly chosen student and μ be the population mean daily screen time of students $H_0 : \mu = 30$ $H_1 : \mu < 30$ Under H_0, since $n = 100$ is large, by Central Limit Theorem, $\bar{X} \sim N \left(30, \frac{\left(\frac{89251}{9900} \right)}{100} \right)$ approximately. Or $Z = \frac{\bar{X} - 30}{\left(\sqrt{\frac{s^2}{n}} \right)} \sim N(0, 1)$ approximately 1 tail z test is used at $\alpha = 0.1$

	Using GC, $p - \text{value} = 0.076055 < 0.1$ Since $p\text{-value} < 0.1$, reject H_0. There is sufficient evidence at the 10% level of significance to show that the population mean screen time per day is less than 30 minutes. Thus Mrs Neha's claim is unjustified.
11 (iii) (c)	$H_0 : \mu = 30$ $H_1 : \mu \neq 30$ Under H_0, since $n = 100$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(30, \frac{\sigma^2}{100}\right)$ approximately, where $Z = \frac{\bar{X} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} \sim N(0,1)$ approximately 2 tail z test is used at $\alpha = 0.05$ Since H_0 is rejected, the test statistic lies inside the critical region (rejection region)  $\frac{30.3 - 30}{\left(\frac{\sigma}{10}\right)} < -1.95996$ or $\frac{30.3 - 30}{\left(\frac{\sigma}{10}\right)} > 1.95996$ $0.3 < -1.95996\left(\frac{\sigma}{10}\right)$ or $0.3 > 1.95996\left(\frac{\sigma}{10}\right)$ $\sigma < -1.53$ (rejected) $0 < \sigma < 1.5306$ $0 < \sigma < 1.53$ (3 s.f) $0 < \sigma^2 < 2.34$